

MATHEMATICAL CALCULATION

Consider the following dataset as a sample used for Random Forest Approach by Boot Strapping Method

Sample	SOC_avg	Temperature (°C)	Cycles	Class
S1	46	18.83	250	Good (1)
S2	46	29.92	249	Good (1)
S3	44	24.91	358	Warning (2)
S4	42	18.96	315	Warning (2)
S5	43	16.13	487	EOL (3)
S6	46	28.51	490	EOL (3)

Now, Consider the first decision tree for SOC_Avg

Sample	SOC_avg	Class
S1	46	Good (1)
S2	46	Good (1)
S3	44	Warning (2)
S4	42	Warning (2)
S5	43	EOL (3)
S6	46	EOL (3)

- 1) Try split between 44 and 46:

Split point:

$$\frac{44 + 46}{2} = 45$$

- 2) Divide the data:

- a. Left Node ($\text{SOC_avg} \leq 45$):

42(2), 43(3), 44(2)

$$\text{Mean}, \bar{y}_L = \frac{2 + 3 + 2}{3} = \frac{7}{3} = 2.33$$

- b. Right Node ($\text{SOC_avg} > 45$):

46(1), 46(1), 46(3)

$$\text{Mean}, \bar{y}_R = \frac{1 + 1 + 3}{3} = \frac{5}{3} = 1.67$$

- 3) Compute MSE:

a. Left Node:

$$\begin{aligned}
 MSE_L &= \frac{(2 - 2.33)^2 + (3 - 2.33)^2 + (2 - 2.33)^2}{3} \\
 &= \frac{(-0.33)^2 + (0.67)^2 + (-0.33)^2}{3} \\
 &= \frac{0.11 + 0.44 + 0.11}{3} = \frac{0.66}{3} = 0.22
 \end{aligned}$$

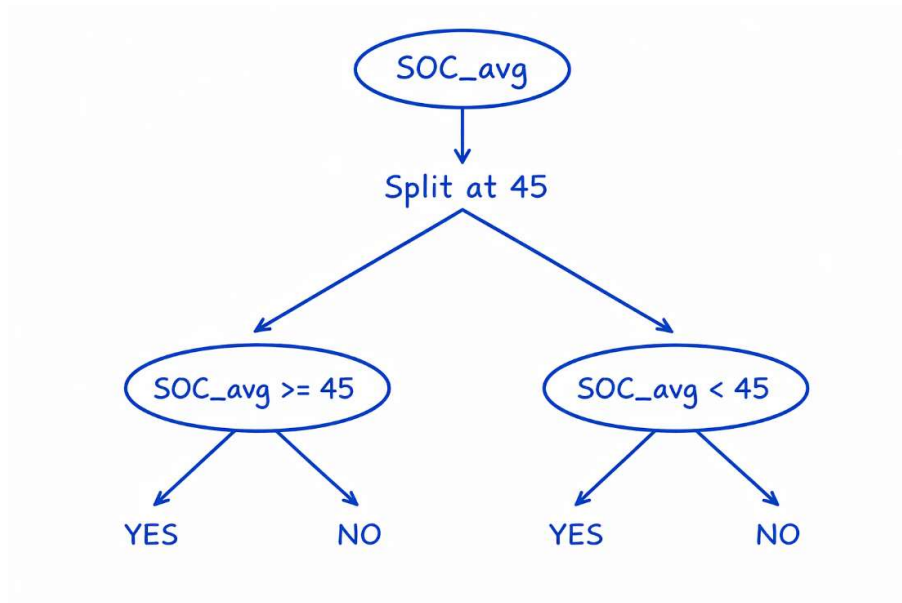
b. Right Node:

$$\begin{aligned}
 MSE_R &= \frac{(1 - 1.67)^2 + (1 - 1.67)^2 + (3 - 1.67)^2}{3} \\
 &= \frac{(-0.67)^2 + (-0.67)^2 + (1.33)^2}{3} \\
 &= \frac{0.44 + 0.44 + 1.77}{3} = \frac{2.65}{3} = 0.88
 \end{aligned}$$

4) Total MSE

$$MSE = MSE_L + MSE_R = 0.22 + 0.88 = 1.10$$

The decision tree is given by



Now, Consider the second decision tree for Temperature

Sample	Temperature (°C)	Class
S1	18.83	Good (1)
S2	29.92	Good (1)
S3	24.91	Warning (2)
S4	18.96	Warning (2)

S5	16.13	EOL (3)
S6	28.51	EOL (3)

- 1) Try split between 24.91 and 28.51:

Split point:

$$\frac{24.91 + 28.51}{2} = 26.71$$

- 2) Divide the data:

- a. Left Node (Temperature ≤ 26.71):

18.83(1), 18.96(2), 16.13(3), 24.91(2)

$$\text{Mean}, \bar{y}_L = \frac{1 + 2 + 3 + 2}{4} = \frac{8}{4} = 2$$

- b. Right Node (Temperature > 26.71):

29.92(1), 28.51(3)

$$\text{Mean}, \bar{y}_R = \frac{1 + 3}{2} = \frac{4}{2} = 2$$

- 3) Compute MSE:

- a. Left Node:

$$\begin{aligned} \text{MSE}_L &= \frac{(1 - 2)^2 + (2 - 2)^2 + (3 - 2)^2 + (2 - 2)^2}{4} \\ &= \frac{(-1)^2 + (0)^2 + (1)^2 + (0)^2}{4} \\ &= \frac{-1 + 0 + 1 + 0}{4} = \frac{0}{4} = 0 \end{aligned}$$

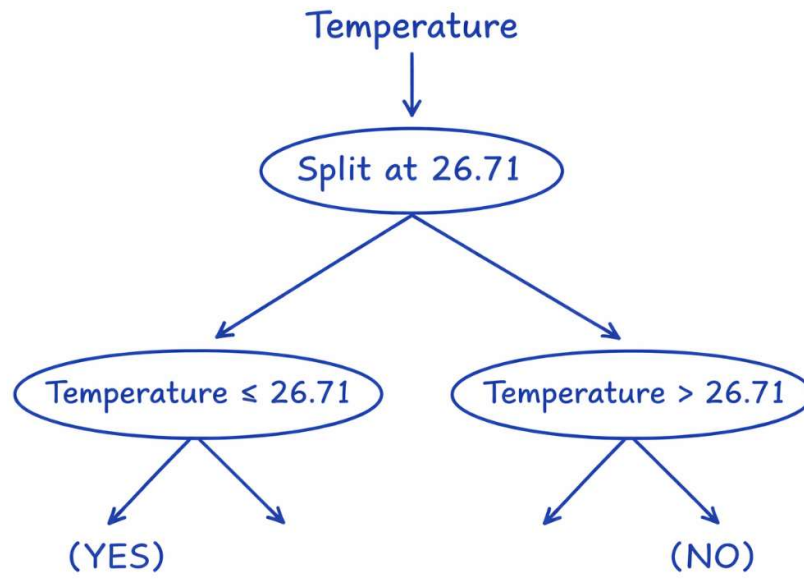
- b. Right Node:

$$\begin{aligned} \text{MSE}_R &= \frac{(1 - 2)^2 + (3 - 2)^2}{2} \\ &= \frac{(-1)^2 + (1)^2}{2} \\ &= \frac{-1 + 1}{2} = \frac{0}{2} = 0 \end{aligned}$$

- 4) Total MSE

$$\text{MSE} = \text{MSE}_L + \text{MSE}_R = 0 + 0 = 0$$

The decision tree is given by



Now, Consider the third decision tree for Cycles

Sample	Cycles	Class
S1	250	Good (1)
S2	249	Good (1)
S3	358	Warning (2)
S4	315	Warning (2)
S5	487	EOL (3)
S6	490	EOL (3)

- 1) Try split between 315 and 358:
Split point:

$$\frac{315 + 358}{2} = 336.5$$

- 2) Divide the data:
 - a. Left Node (Cycle ≤ 336.5):

250(1), 249(1), 315(2)

$$\text{Mean, } \bar{y}_L = \frac{1 + 1 + 2}{3} = \frac{4}{3} = 1.33$$

- b. Right Node (Cycle > 336.5):

358(2), 487(3), 490(3)

$$\text{Mean}, \bar{y}_R = \frac{2 + 3 + 3}{3} = \frac{8}{3} = 2.33$$

3) Compute MSE:

a. Left Node:

$$\begin{aligned} \text{MSE}_L &= \frac{(1 - 1.33)^2 + (1 - 1.33)^2 + (2 - 1.33)^2}{3} \\ &= \frac{(-0.33)^2 + (-0.33)^2 + (0.67)^2}{3} \\ &= \frac{0.11 + 0.11 + 0.44}{3} = \frac{0.66}{3} = 0.22 \end{aligned}$$

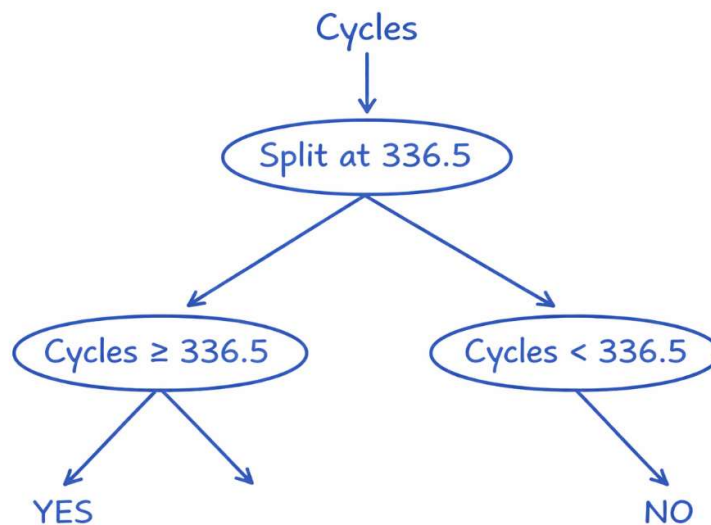
b. Right Node:

$$\begin{aligned} \text{MSE}_R &= \frac{(2 - 2.33)^2 + (3 - 2.33)^2 + (3 - 2.33)^2}{2} \\ &= \frac{(-0.33)^2 + (0.67)^2 + (0.67)^2}{2} \\ &= \frac{0.11 + 0.44 + 0.44}{2} = \frac{0.99}{2} = 0.33 \end{aligned}$$

4) Total MSE

$$\text{MSE} = \text{MSE}_L + \text{MSE}_R = 0.22 + 0.33 = 0.55$$

The decision tree is given by



Hence, The final decision tree is given by

